Supplemental Table 1. Example search strategy performed in Embase (A), Cochrane Registers (B), and PubMed (C). All searches conducted on 04/11/20

#	**A. Searches Performed in Embase**
1	Plastic surgery.mp. or exp plastic surgery/
2	Reconstructive surgical procedures.mp. or exp reconstructive surgery/
3	Esthetics.mp. or exp esthetics/
4	microsurgery.mp. or exp microsurgery/
5	1 or 2 or 3 or 4
6	Artificial Intelligence.mp. or exp artificial intelligence/
7	Algorithms.mp. or exp algorithm/
8	Neural Networks Computer.mp. or exp artificial neural network/
9	Computer Simulation.mp. or exp computer simulation/
10	Machine Learning.mp. or exp machine learning/
11	Deep learning.mp. or exp deep learning/
12	Pattern Recognition, Automated.mp. or exp automated pattern recognition/
13	Three-Dimensional Imaging.mp. or exp three dimensional imaging/
14	Automation.mp. or exp automation/
15	Electronic Data Processing.mp. or exp information processing/
16	Computer-Aided Design.mp. or exp computer aided design/
17	Computer Security.mp. or exp computer security/
18	Computer Simulation.mp. or exp computer simulation/
19	Image Processing, Computer-Assisted.mp. or exp image processing/
20	Signal Processing, Computer-Assisted.mp. or exp signal processing/
21	Facial Recognition.mp. or exp facial recognition/
22	Biomedical Technology.mp. or exp medical technology/
23	Computer assisted diagnosis.mp. or exp computer assisted diagnosis/
24	Robotics.mp. or exp robotics/
25	Electronic Data Processing.mp. or exp information processing/
26	Big data.mp. or exp big data/

27	Radiographic Image Interpretation, Computer Assisted.mp. or exp computer assisted diagnosis/
28	Biometric Identification.mp. or exp biometry/
29	6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28
30	5 and 29
31	limit 30 to english language
32	6 or 8 or 10 or 11
33	31 and 32
#	**B. Searches Performed in Cochrane Registers**
1	Plastic surgery.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
2	Reconstructive surgical procedures.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
3	Esthetics.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
4	microsurgery.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
5	1 or 2 or 3 or 4
6	Artificial Intelligence.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
7	Algorithms.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
8	Neural Networks Computer.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
9	artificial neural network.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
10	Computer Simulation.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
11	Machine Learning.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
12	Deep learning.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
13	Pattern Recognition, Automated.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
14	Three-Dimensional Imaging.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
15	Automation.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
16	Electronic Data Processing.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
17	Information processing.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
18	Computer-Aided Design.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
19	Computer Security.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
20	Computer Simulation.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
21	Image Processing, Computer-Assisted.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]

22	Signal Processing, Computer-Assisted.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
23	Facial Recognition.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
24	Biomedical Technology.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
25	Computer assisted diagnosis.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
26	Robotics.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
27	Electronic Data Processing.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
28	Information processing.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
29	Big data.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
30	Radiographic Image Interpretation, Computer Assisted.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
31	Computer assisted diagnosis.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
32	Biometric Identification.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
33	biometry.mp. [mp=ti, ab, tx, kw, ct, ot, sh, hw]
34	6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33
35	5 and 34
36	limit 35 to english language [Limit not valid in CCA,CDSR,CLCMR; records were retained]

#	**C. Searches Performed in PubMed**
1	(((ai artificial intelligence[MeSH Terms])) OR (ais artificial intelligence[MeSH Terms])) AND ((english[Language]) AND ((((((((((((((((((((((plastic surgery[MeSH Terms])) OR (neck plastic surgeries[MeSH Terms])) OR (neck plastic surgery[MeSH Terms])) OR (plastic surgeries, neck[MeSH Terms])) OR (plastic surgery, neck[MeSH Terms])) OR (cosmetic reconstructive surgical procedures[MeSH Terms])) OR (procedure, reconstructive surgical[MeSH Terms])) OR (procedures, reconstructive surgical[MeSH Terms])) OR (reconstructive surgical procedure[MeSH Terms])) OR (reconstructive surgical procedures[MeSH Terms])) OR (breast reconstruction[MeSH Terms])) OR (breast reconstructions[MeSH Terms])) OR (cosmetic reconstructive surgical procedures[MeSH Terms])) OR (esthetics[MeSH Terms])) OR (cerebral microsurgical revascularization[MeSH Terms])) OR (cerebral microsurgical revascularizations[MeSH Terms])) OR (flap, microsurgical free[MeSH Terms])) OR (flaps, microsurgical free[MeSH Terms])) OR (free flap, microsurgical[MeSH Terms])) OR (microsurgery[MeSH Terms])) AND ((ai artificial intelligence[MeSH Terms]) OR (ais artificial intelligence[MeSH Terms])) OR (artificial intelligence[MeSH Terms])) OR (algorithm[MeSH Terms])) OR (algorithms[MeSH Terms])) OR (model, neural network[MeSH Terms])) OR (models, neural network[MeSH Terms])) OR (neural network anatomic[MeSH Terms])) OR (neural network computer[MeSH Terms])) OR (neural network model[MeSH Terms])) OR (computer simulation[MeSH Terms])) OR (computer simulations[MeSH Terms])) OR (machine learning[MeSH Terms])) OR (deep learning[MeSH Terms])) OR (artificial neural network[MeSH Terms])) OR (automated pattern recognition[MeSH Terms])) OR (3d imaging, computer generated[MeSH Terms])) OR (3d

	imagings, computer generated[MeSH Terms])) OR (computer assisted three dimensional imaging[MeSH Terms])) OR (computer assisted three dimensional imagings[MeSH Terms])) OR (image, three dimensional[MeSH Terms])) OR (automation[MeSH Terms])) OR (automations[MeSH Terms])) OR (electronic data processing[MeSH Terms])) OR (automatic information processing[MeSH Terms])) OR (information processing, automatic[MeSH Terms])) OR (computer aided design[MeSH Terms])) OR (computer aided designs[MeSH Terms])) OR (computer aided manufacturing[MeSH Terms])) OR (computer aided surgeries[MeSH Terms])) OR (computer aided surgery[MeSH Terms])) OR (analysis, computer assisted image[MeSH Terms])) OR (computer security[MeSH Terms])) OR (computer simulation[MeSH Terms])) OR (computer simulations[MeSH Terms])) OR (analysis, computer assisted image[MeSH Terms])) OR (computer assisted image analyses[MeSH Terms])) OR (computer assisted image analysis[MeSH Terms])) OR (computer assisted image interpretation[MeSH Terms])) OR (computer assisted image interpretations[MeSH Terms])) OR (computer assisted signal interpretation[MeSH Terms])) OR (computer assisted signal interpretations[MeSH Terms])) OR (computer assisted signal processing[MeSH Terms])) OR (interpretation, computer assisted signal[MeSH Terms])) OR (interpretations, computer assisted signal[MeSH Terms])) OR (automated facial recognition[MeSH Terms])) OR (automated facial recognitions[MeSH Terms])) OR (assessment, biomedical technology[MeSH Terms])) OR (assessments, biomedical technology[MeSH Terms])) OR (computer assisted diagnoses[MeSH Terms])) OR (computer assisted diagnosis[MeSH Terms])) OR (robotics[MeSH Terms])) OR (operation, remote robotics[MeSH Terms])) OR (operations, remote robotics[MeSH Terms])) OR (remote operation robotics[MeSH Terms])) OR (remote operations robotics[MeSH Terms])) OR (electronic data processing[MeSH Terms])) OR (processing, electronic data[MeSH Terms])) OR (automatic information processing[MeSH Terms])) OR (information processing[MeSH Terms])) OR (information processing, automatic[MeSH Terms])) OR (big data[MeSH Terms])) OR (computer assisted radiographic image interpretation[MeSH Terms])) OR (radiographic image interpretation, computer assisted[MeSH Terms])) OR (authentication, biometric[MeSH Terms])) OR (authentications, biometric[MeSH Terms])) OR (biometric authentication[MeSH Terms])) OR (biometric authentications[MeSH Terms])) OR (biometric identification[MeSH Terms])) OR (biometry[MeSH Terms]))))